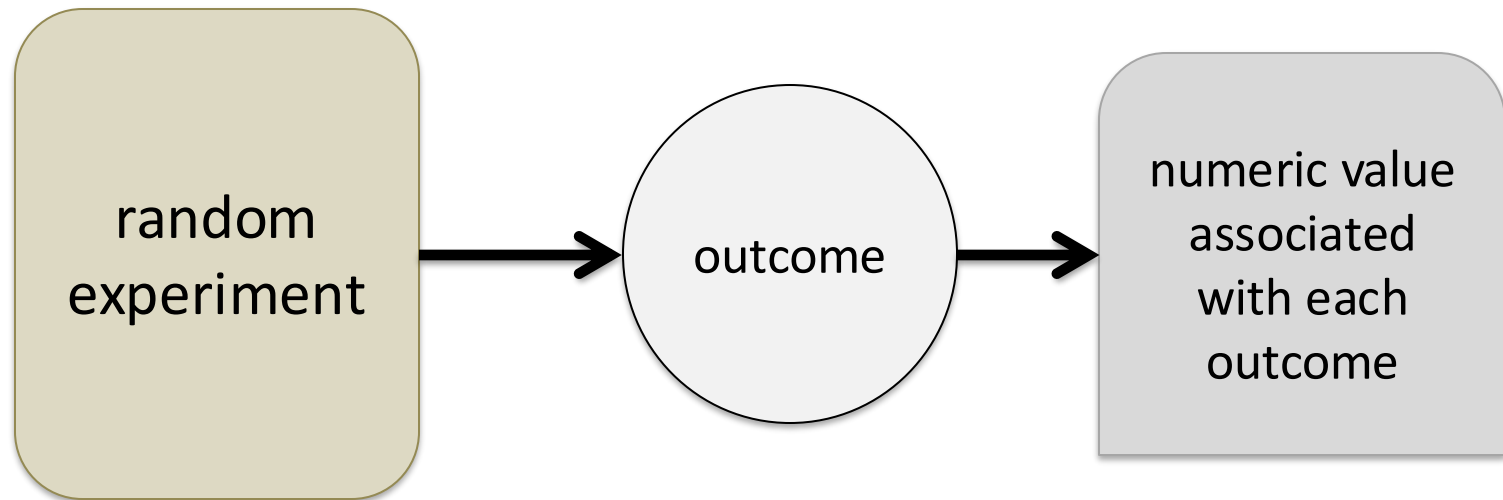


IEOR 4101: Probability, Statistics and Simulation

Lecture 5: (Discrete) Random Variables and Expectation

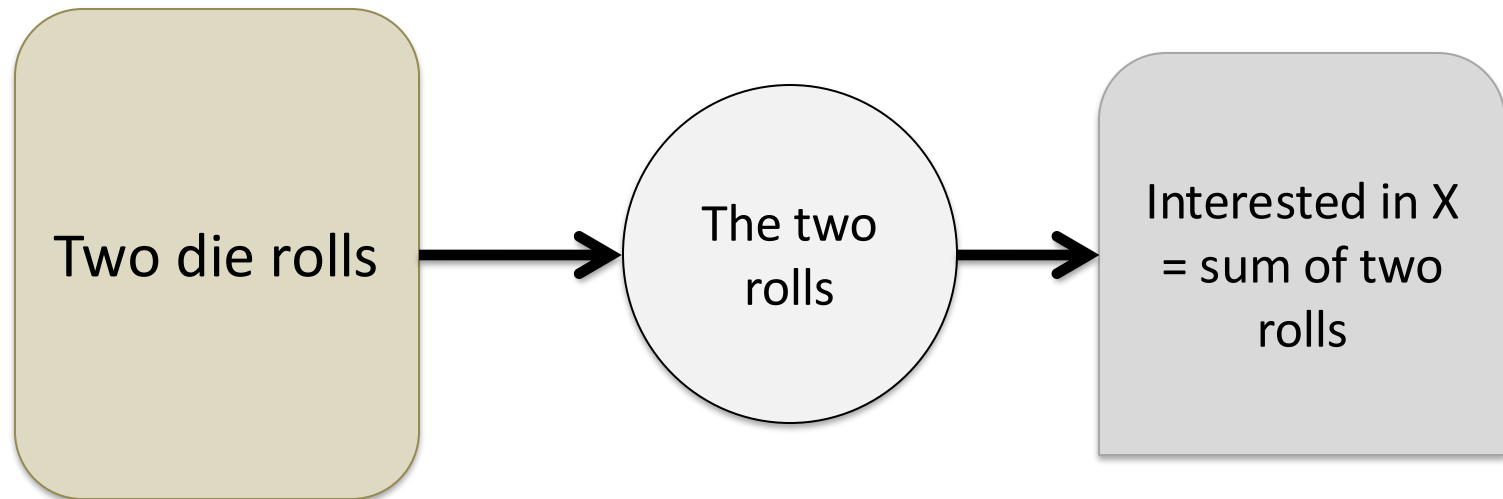
Random Variables

- Random variable summarizes or represents the outcomes with numerical values



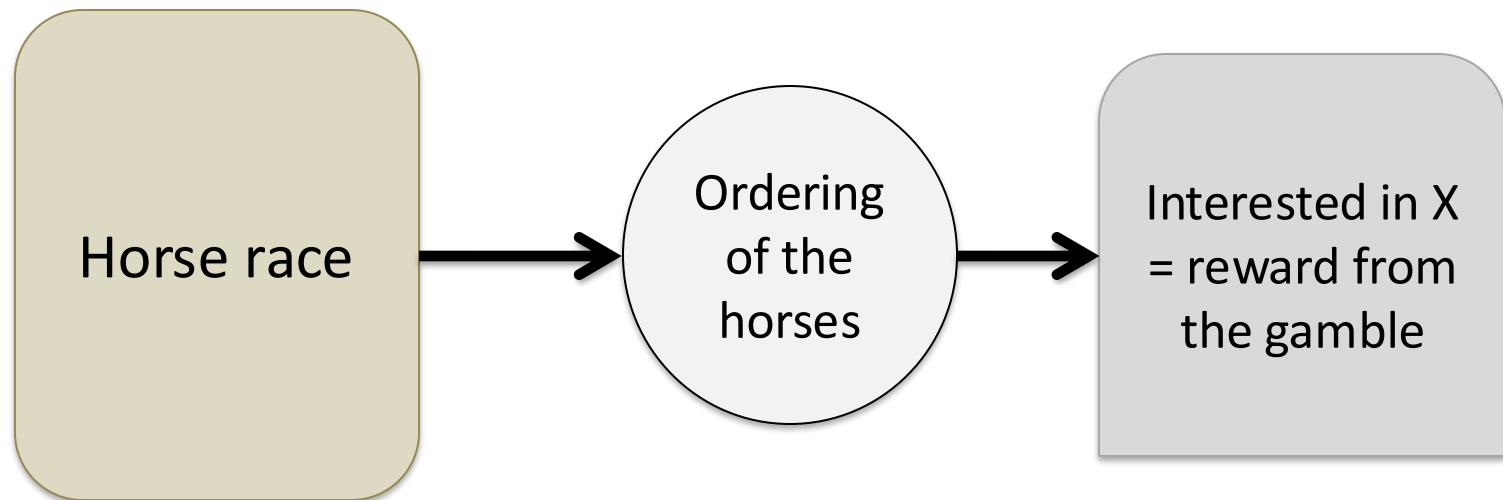
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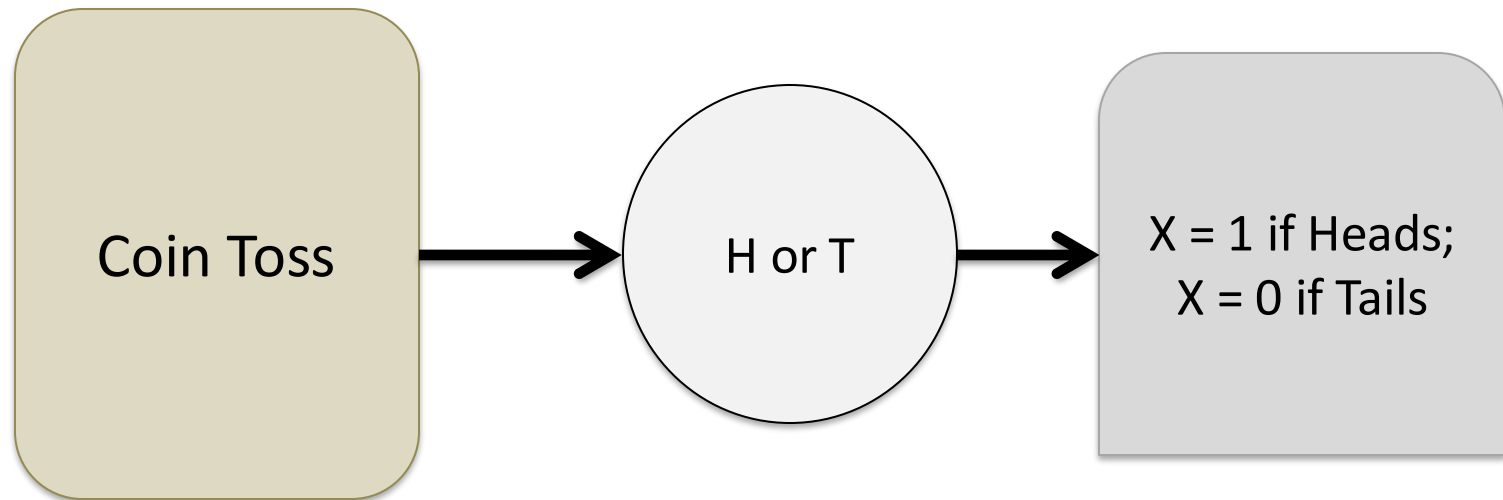
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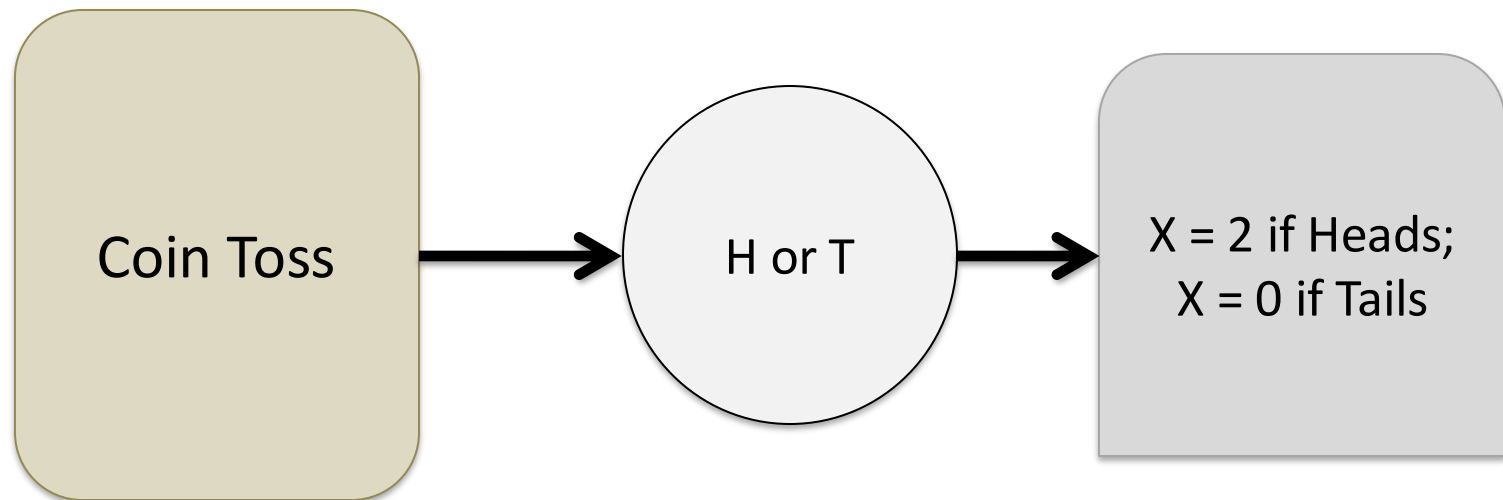
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- Random variable summarizes or represents the outcomes with numerical values
- These assignments of numerical values may be different, even for the same outcome, depending on interest



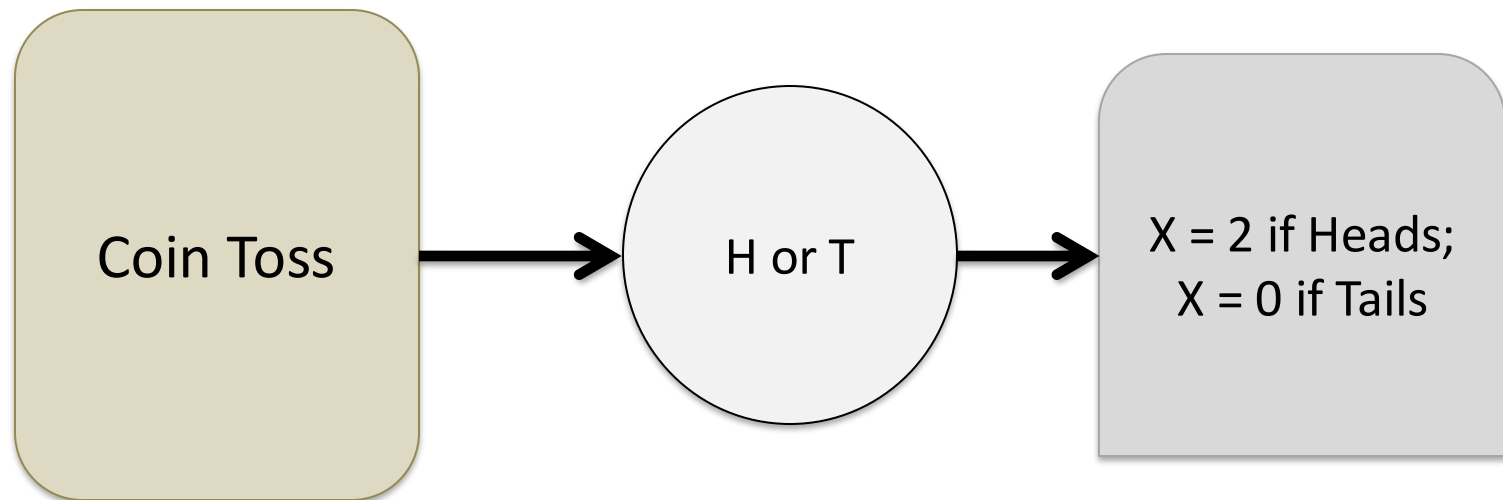
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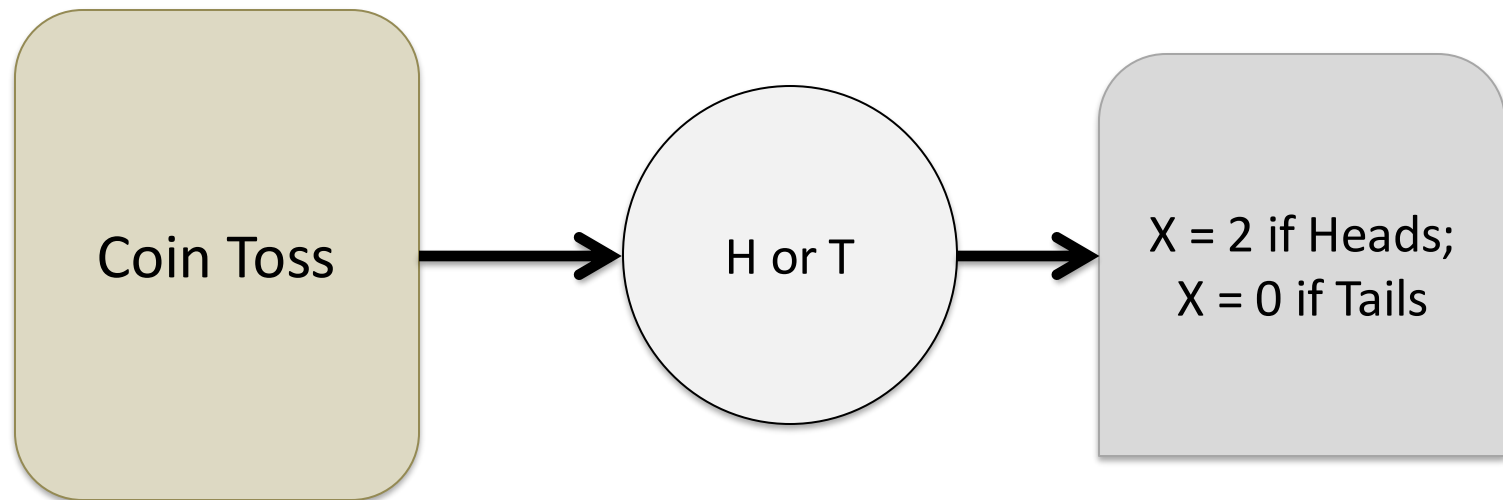
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- The realized value of a random variable is random



Random Variables

- Random variable summarizes or represents the outcomes with numerical values
- These assignments of numerical values may be different, even for the same outcome, depending on interest
- The realized value of a random variable is random
- To specify a random variable, identify its range of values and probability assigned to each value



Random Variables

- **Example.** I purchased two electronic components, each of which may be defective or acceptable
- $(d, d), (d, a), (a, d), (a, a)$ – probabilities 0.09, 0.21, 0.21 and 0.49 respectively
- Let X = number of acceptable components; X can be 0, 1, or 2

$$P(X = 0) = P((d, d)) = 0.09$$

$$P(X = 1) = P((d, a) \text{ or } (a, d)) = P(d, a) + P(a, d)$$

$$P(X = 2) = P((a, a)) = 0.49$$

$0.21 + 0.21$
 0.42

Random Variables

- **Example.** I purchased two electronic components, each of which may be defective or acceptable
- $(d, d), (d, a), (a, d), (a, a)$ – probabilities 0.09, 0.21, 0.21 and 0.49 respectively
- Let X = number of acceptable components; X can be 0, 1, or 2
- $P(X = 0) = 0.09$; $P(X = 1) = 0.42$; $P(X = 2) = 0.49$

Random Variables

- **Example.** I purchased two electronic components, each of which may be defective or acceptable
- $(d, d), (d, a), (a, d), (a, a)$ – probabilities 0.09, 0.21, 0.21 and 0.49 respectively
- Let X = number of acceptable components; X can be 0, 1, or 2
- $P(X = 0) = 0.09$; $P(X = 1) = 0.42$; $P(X = 2) = 0.49$
- Let I = indicator whether there is *at least* one acceptable component

$$I = \begin{cases} 1 & \text{if at least 1 acceptable} \\ 0 & \text{otherwise} \end{cases}$$

$$P(I = 1) = 0.21 + 0.21 + 0.49 = 0.91 \quad P(I = 0) = 0.09$$

Random Variables

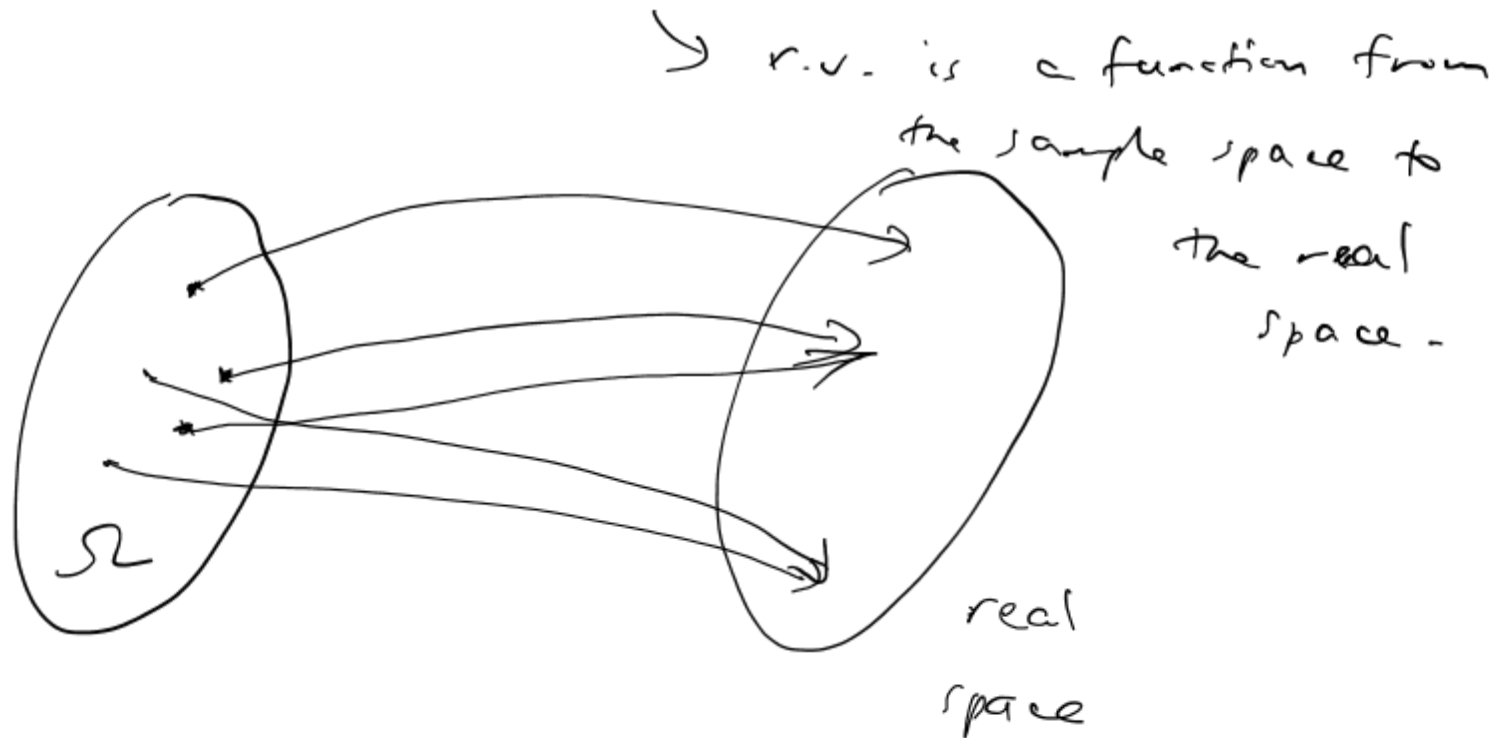
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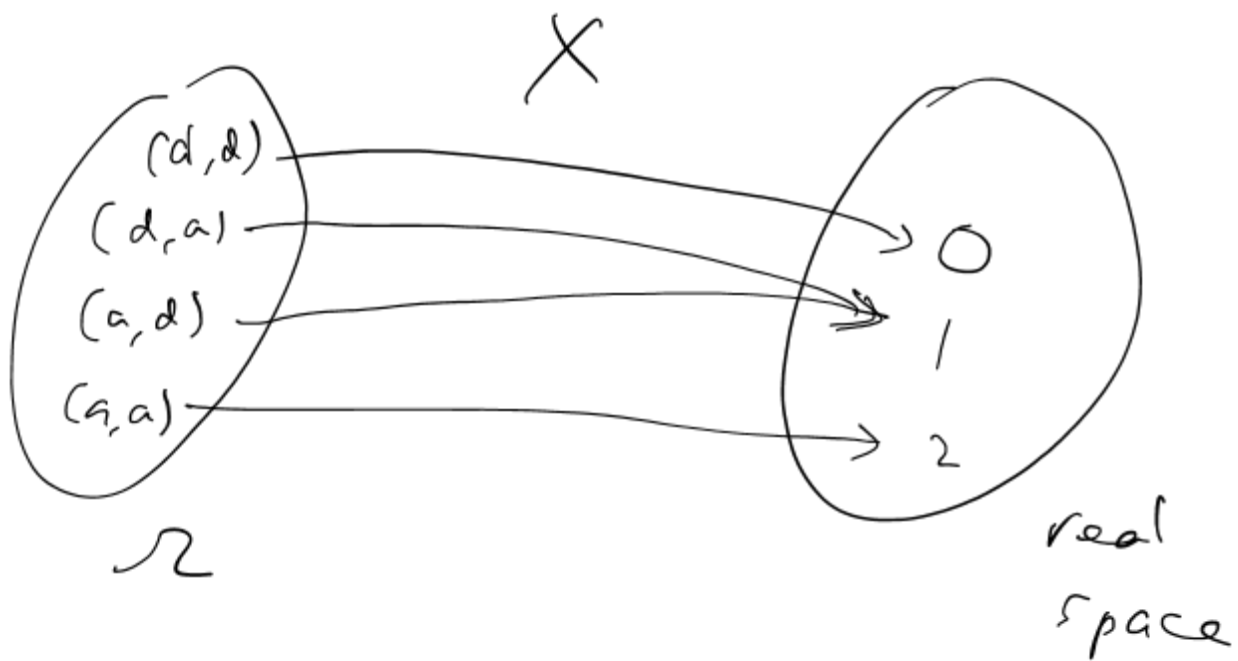
Random Variables

- In summary: a *random variable* assigns numerical values to outcomes of the sample space

Random Variables

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- More formally: a *random variable* is a *function* that assigns a numerical value to each outcome in the sample space





Random Variables

- In summary: a *random variable* assigns numerical values to outcomes of the sample space
- More formally: a *random variable* is a *function* that assigns a numerical value to each outcome in the sample space
- A random variable is *discrete* if the range of values that it takes is finite or “can be enumerated”
- Possible range of values:
 - $\{0, 1\}$ (coin toss example)
 - $\{0.5, 2, \pi, e\}$
 - $\{0, 1, 2, 3, 4, \dots\}$

Random Variables

- *Probability mass function* (pmf) $p(a)$ of a discrete random variable X is $p(a) = P(X = a)$, for each possible value a of X
- $p(a)$ can be positive for at most a *countable* number of values of a – if X takes values $x_1, x_2, x_3, \dots$, then $p(x_i) > 0, i = 1, 2, \dots$ and $p(x) = 0$ for all other values of x
- **Example.** Suppose a company hired 6 employees recently and let X be the number of females in the group. Further suppose that $P(X = 1) = 0.1, P(X = 2) = 0.3, P(X = 3) = 0.5, P(X = 4) = 0.05, P(X = 5) = P(X = 6) = 0.025$

pmf of X :

$$p(1) = 0.1$$

$$p(2) = 0.3$$

$$p(3) = 0.5$$

$$p(4) = 0.05$$

$$p(5) = 0.025$$

$$p(6) = 0.025$$

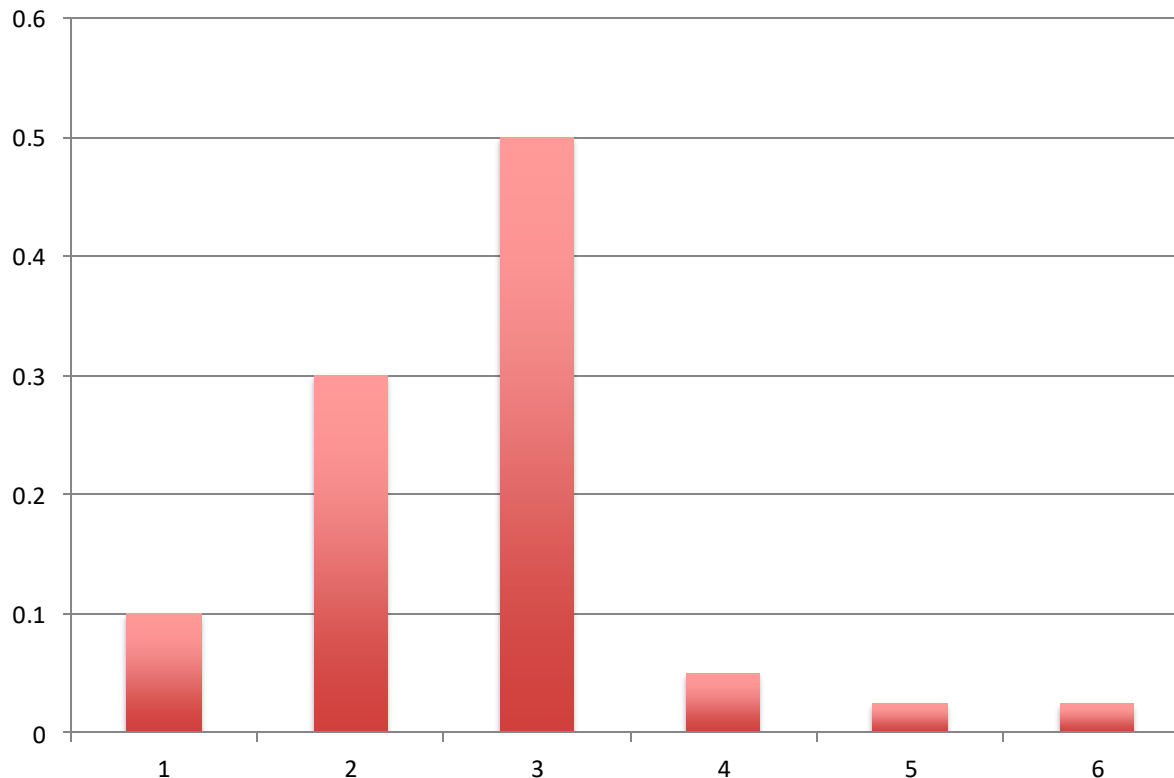
$$p(x) = 0 \text{ for all other } x.$$

Random Variables

- *Probability mass function* (pmf) $p(a)$ of a discrete random variable X is $p(a) = P(X = a)$, for each possible value a of X
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- **Example.** Suppose a company hired 6 employees recently and let X be the number of females in the group. Further suppose that $P(X = 1) = 0.1$, $P(X = 2) = 0.3$, $P(X = 3) = 0.5$, $P(X = 4) = 0.05$, $P(X = 5) = P(X = 6) = 0.025$
- The pmf of X is specified by $p(1) = 0.1$, $p(2) = 0.3$, $p(3) = 0.5$, $p(4) = 0.05$, $p(5) = p(6) = 0.025$, and $p(x) = 0$ for all other values of x

Random Variables

- Here is the plot of the pmf
- Looks a lot like histogram – indeed they are very similar ideas



Random Variables

- Can also define the *cumulative distribution function* (cdf) of a random variable X as $F(x) = P(X \leq x) = \sum_{a \leq x} P(X = a) = \sum_{a \leq x} p(a)$

X has pmf:

$$p(1) = 0.1, \quad p(2) = 0.3, \quad p(3) = 0.5, \quad p(4) = 0.05$$

$$p(5) = 0.025, \quad p(6) = 0.025$$

$$F(1) = \sum_{a \leq 1} p(a) = p(1) = 0.1$$

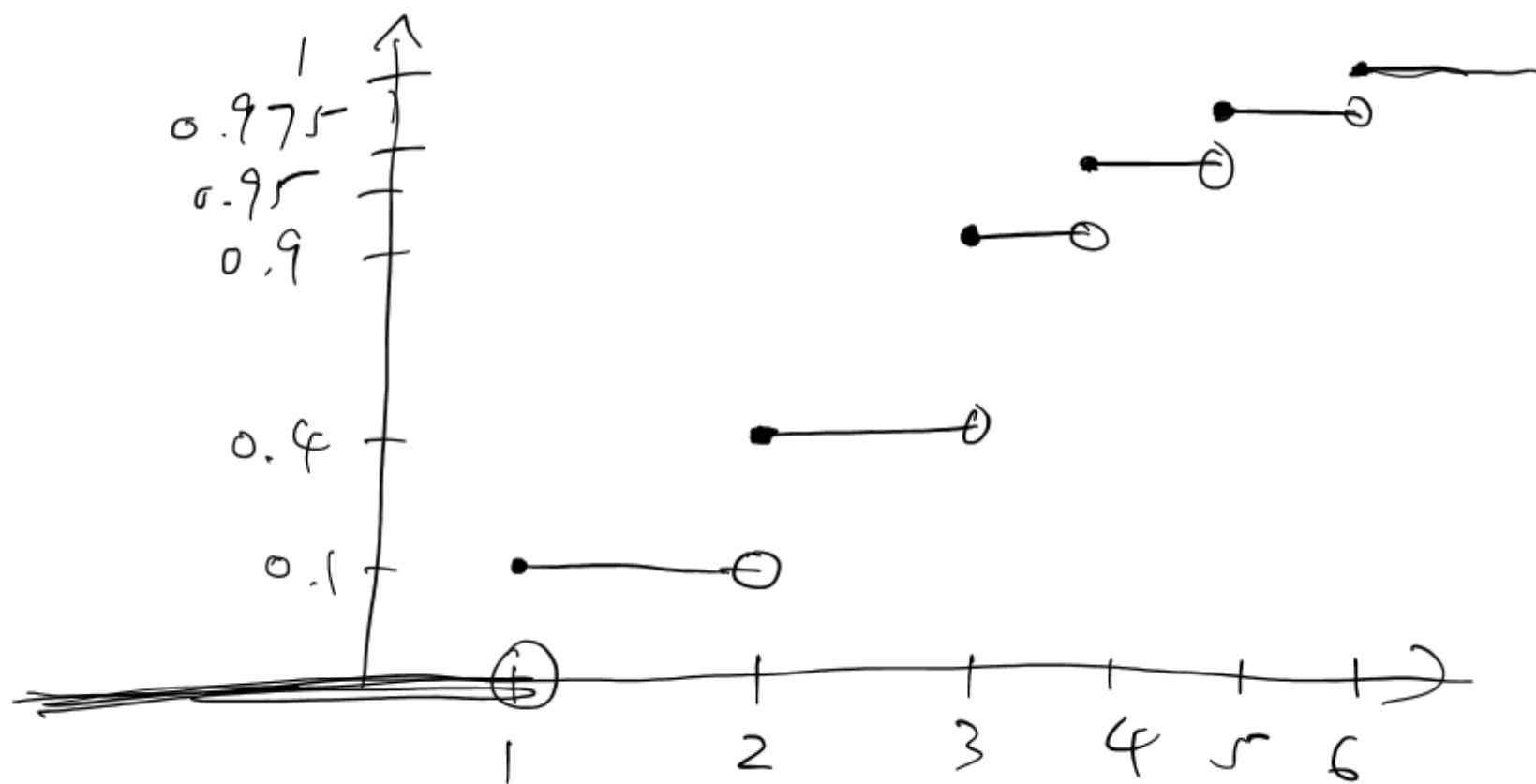
$$F(2) = p(1) + p(2) = 0.4$$

$$F(2.5) = p(1) + p(2) = 0.4$$

$$F(100) = p(1) + \dots + p(6) = 1$$

$$F(0.1) = 0 \quad F(-100) = 0$$

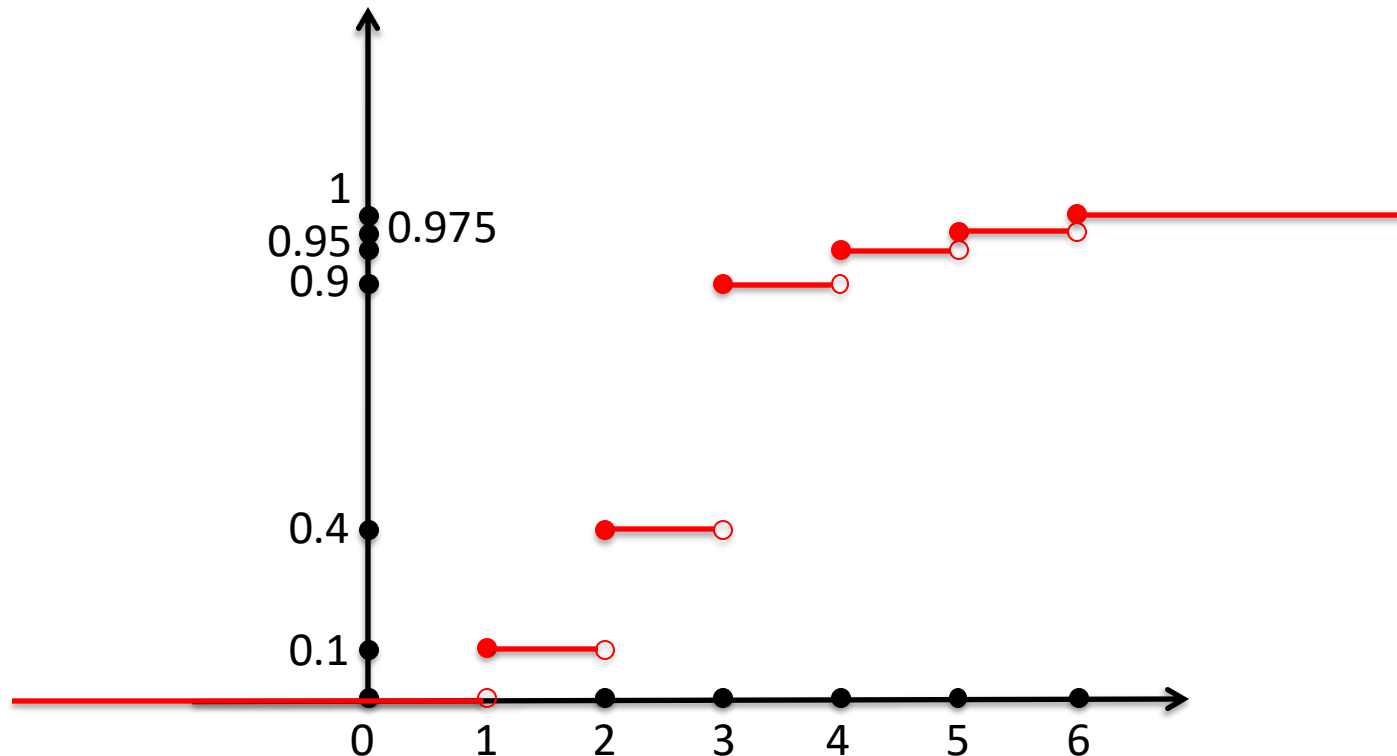
$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ 0.1 & \text{if } 1 \leq x < 2 \\ 0.4 & \text{if } 2 \leq x < 3 \\ 0.9 & \text{if } 3 \leq x < 4 \\ 0.95 & \text{if } 4 \leq x < 5 \\ 0.975 & \text{if } 5 \leq x < 6 \\ 1 & \text{if } x \geq 6 \end{cases}$$



Random Variables

- Can also define the *cumulative distribution function* (cdf) of a random variable X as $F(x) = P(X \leq x) = \sum_{a \leq x} P(X = a) = \sum_{a \leq x} p(a)$
- The cdf of X in the previous example:

$F(x) = 0$ for $x < 1$, 0.1 for $1 \leq x < 2$, 0.4 for $2 \leq x < 3$, 0.9 for $3 \leq x < 4$, 0.95 for $4 \leq x < 5$, 0.975 for $5 \leq x < 6$, and 1 for $x \geq 6$



Properties:

- step function, i.e. it stays constant and jump, and stay constant again etc.
- starts from 0 at $-\infty$ and ends at 1 at ∞
- non-decreasing
- when there is a jump, the "solid" circle is in the top line

Random Variables

- The pmf/cdf of a random variable X specifies it completely
- Knowing pmf is equivalent to knowing cdf, which is equivalent to specifying X completely

From the cdf in the example,
we see $1, 2, \dots, 6$ has jumps.

$\Rightarrow$ these are the possible values X can take.

For each of $1, 2, \dots, 6$,

jump size = pmf.

e.g. $p(1) = 0.1$

$$p(2) = 0.4 - 0.1 = 0.3 \quad \dots$$

Random Variables

- We are sometimes interested in the joint behavior of more than one random variable
- The *joint probability mass function* of two (discrete) random variables X and Y is $p(x,y) = P(X=x, Y=y)$

Random Variables

- **Example.** In a survey conducted by AIG, investors were asked to rate how knowledgeable they felt they were as investors. Both online and traditional investors were included in the survey. The following data were resulted:
- Of the online investors: 8% - savvy, 55% - experienced and 37% - novice
- Of the traditional investors: 4% - savvy, 29% - experienced and 67% - novice
- $\frac{1}{3}$ of the surveyed population were traditional
- Pick one at random from the population

$$X = \begin{cases} 0 & \text{if online} \\ 1 & \text{if traditional} \end{cases}$$

$$Y = \begin{cases} 0 & \text{if savvy} \\ 1 & \text{if experienced} \\ 2 & \text{if novice} \end{cases}$$

joint pmf :

	$Y=0$	$Y=1$	$Y=2$
$X=0$	$\frac{2}{3} \times 8\%$	$\frac{2}{3} \times 55\%$	$\frac{2}{3} \times 37\%$
$X=1$	$\frac{1}{3} \times 4\%$	$\frac{1}{3} \times 29\%$	$\frac{1}{3} \times 67\%$

$$p(0,0) = \frac{2}{3} \times 8\%, \dots$$

Random Variables

- Two random variables are of interest: whether the picked person is traditional or online; what was his/her response
- Let $X = 0$ if person is online and $X = 1$ if person is traditional
- Let $Y = 2$ if responded savvy, $Y = 1$ if responded experienced, and $Y = 0$ if responded novice
- Each entry in the table shows $P(X=a, Y=b)$, e.g., $P(X=0, Y=0) = 37\% \times 2/3$

	$Y = 2$ (savvy)	$Y = 1$ (experienced)	$Y = 0$ (novice)
$X = 0$ (online)	$8\% \times 2/3$	$55\% \times 2/3$	$37\% \times 2/3$
$X = 1$ (traditional)	$4\% \times 1/3$	$29\% \times 1/3$	$67\% \times 1/3$

Random Variables

- Given the joint pmf of two (discrete) random variables X and Y is $p(x,y) = P(X=x, Y=y)$, the marginal pmfs are given by

$$p_X(x) = P(X=x) = \sum_y P(X=x, Y=y) = \sum_y p(x,y)$$

and

$$p_Y(y) = P(Y=y) = \sum_x P(X=x, Y=y) = \sum_x p(x,y)$$

Random Variables

- What are the marginal pmfs of X and Y?

	Y = 2 (savvy)	Y = 1 (experienced)	Y = 0 (novice)
X = 0 (online)	$8\% \times 2/3$	$55\% \times 2/3$	$37\% \times 2/3$
X = 1 (traditional)	$4\% \times 1/3$	$29\% \times 1/3$	$67\% \times 1/3$

$$P_X(0) = 8\% \times \frac{2}{3} + 55\% \times \frac{2}{3} + 37\% \times \frac{2}{3} = \frac{2}{3}$$

$$P_X(1) = 4\% \times \frac{1}{3} + 29\% \times \frac{1}{3} + 67\% \times \frac{1}{3} = \frac{1}{3}$$

$$P_Y(2) = 8\% \times \frac{2}{3} + 4\% \times \frac{1}{3}, \quad P_Y(1) = 55\% \times \frac{2}{3} + 29\% \times \frac{1}{3}, \quad P_Y(0) = 37\% \times \frac{2}{3} + 67\% \times \frac{1}{3}$$

Random Variables

- What are the marginal pmfs of X and Y?
- Row sums give the marginal pmf of X and column sums give pmf of Y

	Y = 2 (savvy)	Y = 1 (experienced)	Y = 0 (novice)
X = 0 (online)	$8\% \times 2/3$	$55\% \times 2/3$	$37\% \times 2/3$
X = 1 (traditional)	$4\% \times 1/3$	$29\% \times 1/3$	$67\% \times 1/3$

$$P(Y=2) = P(X=0, Y=2) + P(X=1, Y=2)$$

Random Variables

- Definition: two discrete random variables X and Y are *independent* if for any x and y , $P(X=x, Y=y) = P(X=x)P(Y=y)$
- In other words, $p(x, y) = p_X(x)p_Y(y)$ for all x and y
- **Example.** Are X and Y in our previous example independent?

We need to check whether

$$P(x, y) = P_X(x) P_Y(y) \quad \text{for all } x, y.$$

0, 1, 2 0, 1, 2

$$P(0, 2) = 8\% = \frac{2}{3}$$

$$P_X(0) = \frac{2}{3}$$

$$P_Y(2) = 8\% \times \frac{2}{3} + 4\% \times \frac{1}{3}$$

$$P(0, 2) \neq P_X(0) P_Y(2)$$

So X & Y are
not indep.

Expectation of Random Variables

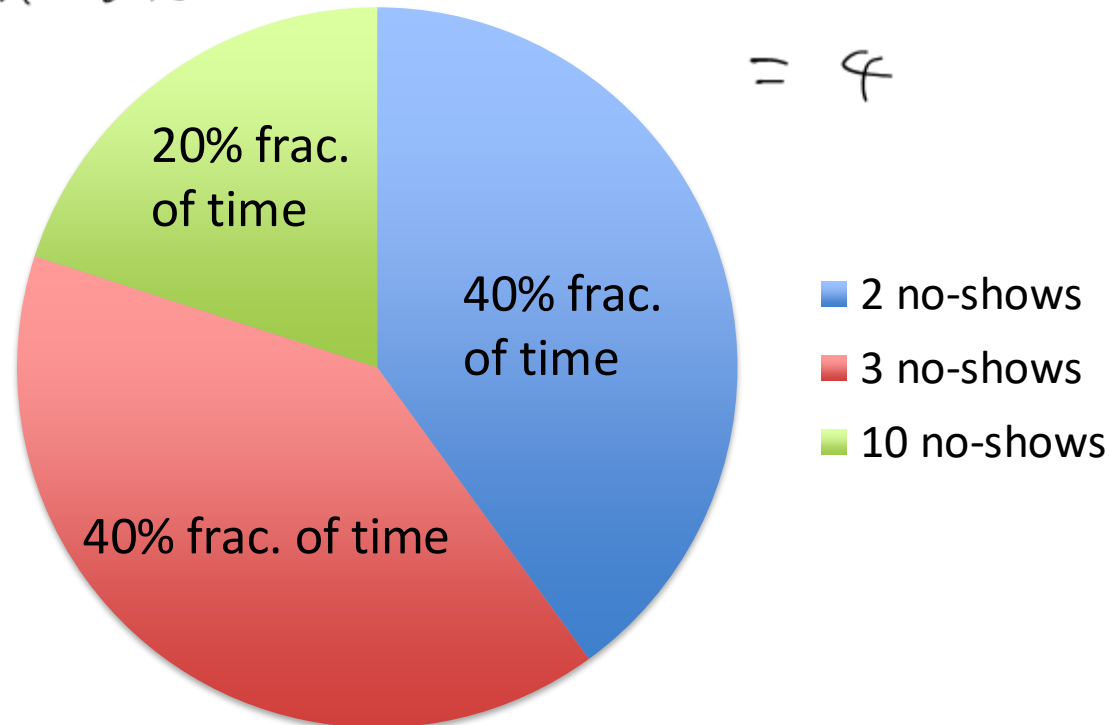
- Arguably the most important characteristic of a random variable is its *expected value/mean/expectation*
- Definition: the expected value of a discrete random variable X is defined by $E[X] = \sum_x xP(X = x) = \sum_x xp(x)$

Expectation of Random Variables

- Airline overbooking example
- Let X = # of no-shows

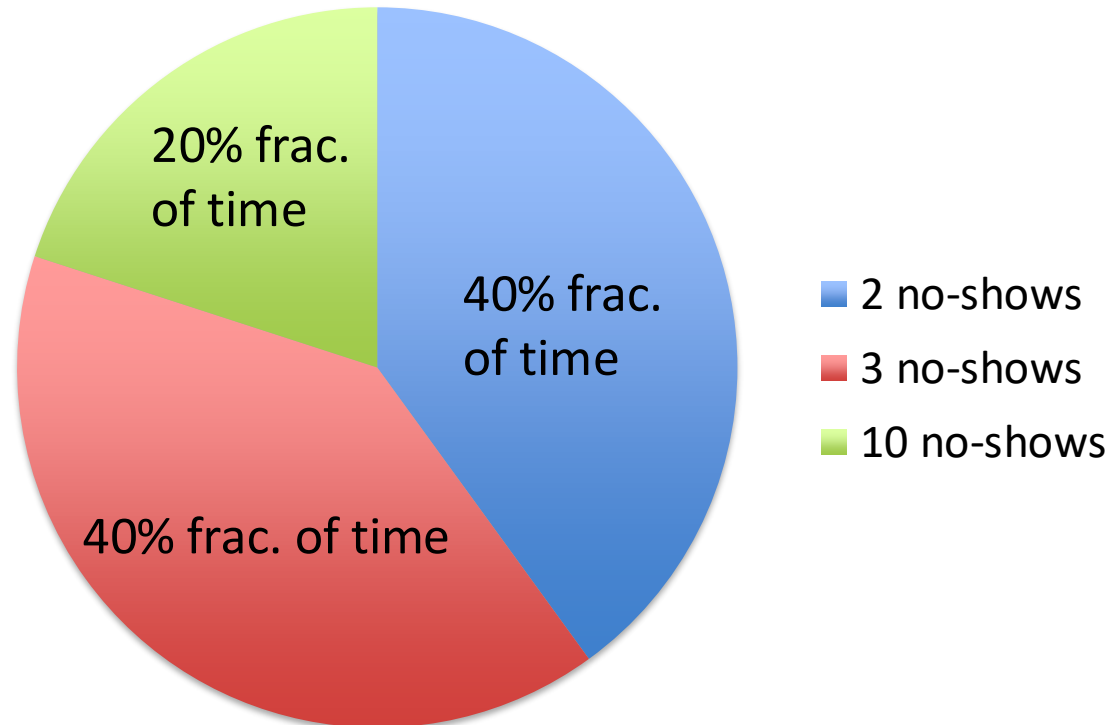
pmf of X : $p(2) = 0.4$, $p(3) = 0.4$, $p(10) = 0.2$
 $p(x) = 0$ for all other x .

expected # no-shows = $E[X] = 2 \times 0.4 + 3 \times 0.4 + 10 \times 0.2$
 $= 4$



Expectation of Random Variables

- Airline overbooking example
- Let X = # of no-shows
- $p(2) = P(X=2) = 40\%$, $p(3) = 40\%$ and $p(10) = 20\%$
- $E[X] = 2 \times p(2) + 3 \times p(3) + 10 \times p(10) = 4$



Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?

X = outcome of die roll

X takes values $1, 2, \dots, 6$

pmf: $p(1) = \dots = p(6) = \frac{1}{6}$

and $p(x) = 0$ for all other x .

$$E(X) = 1 \times \frac{1}{6} + \dots + 6 \times \frac{1}{6} = 3.5$$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?
- $E[X] = 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?
- $E[X] = 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$
- **Example.** What is the expected value of the number of coin tosses until we see a Heads, for a biased coin of coming up Heads with probability 0.3?

X = # toss to see a H

X take values $1, 2, 3, \dots$

$$p(1) = 0.3$$

$$p(k) = 0.7^{k-1} \cdot 0.3$$

$$p(2) = 0.7 \times 0.3$$

$$p(3) = 0.7^2 \times 0.3$$

$$\begin{aligned}
 E(x) = & 1 \times 0.3 + 2 \times 0.7 \times 0.3 \\
 & + 3 \times 0.7^2 \times 0.3 \\
 & \vdots \\
 & + k \times 0.7^{k-1} \times 0.3 \\
 & \vdots
 \end{aligned}$$

$$= \dots$$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?
- $E[X] = 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$
- **Example.** What is the expected value of the number of coin tosses until we see a Heads, for a biased coin of coming up Heads with probability 0.3?
- Let X = the number of tosses until Heads. Then $P(X=1)=0.3$, $P(X=2)=0.7 \times 0.3$, ..., $P(X=k) = 0.7^{k-1} \times 0.3$, ...
- $E[X] = 1 \times P(X=1) + 2 \times P(X=2) + 3 \times P(X=3) + \dots = \sum_k k \times 0.7^{k-1} \times 0.3 = \dots = 1/0.3 = 10/3$

Expectation of Random Variables

- **Example.** Pick two distinct number at random from 1, 2, ..., 100. What is the expected value of the larger one?

X = larger number between the 2 distinct numbers

X take values 2, 3, 4, ..., 100

$$p(2) = P(X=2)$$

$$= P(1^{\text{st}} \# = 1, 2^{\text{nd}} \# = 2)$$

$$+ P(1^{\text{st}} \# = 2, 2^{\text{nd}} \# = 1)$$

$$= P(1^{\text{st}} \# = 1) \cdot P(2^{\text{nd}} \# = 2 | 1^{\text{st}} \# = 1) +$$

$$P(1^{\text{st}} \neq 2) P(2^{\text{nd}} \neq 1 \mid 1^{\text{st}} \neq 2)$$

$$= \frac{1}{100} \times \frac{1}{99} + \frac{1}{100} \times \frac{1}{99}$$

$$= 2 \times \frac{1}{100} \times \frac{1}{99}$$

$$P(k) = P(X = k) = 2 \times \frac{1}{100} \times \frac{k-1}{99}$$

$$E[X] = \sum_{k=2}^{100} k \times 2 \times \frac{1}{100} \times \frac{k-1}{99}$$

Expectation of Random Variables

- **Example.** Pick two distinct number at random from 1, 2, ..., 100. What is the expected value of the larger one?
- Let X = the larger number. Event $\{X=k\}$ is the event that number k is picked, and another one is from 1, 2, ..., $k-1$.
- $P(X=k) = 2 \times 1/100 \times (k-1)/99 = 2 \times (k-1)/(100 \times 99)$
- $E[X] = 2 \times P(X=2) + 3 \times P(X=3) + \dots + 100 \times P(X=100)$

Sum of Expectations

- Often interested in sums of random variables
- E.g., quarterly earnings X and Y – interested in $E[X+Y]$
- E.g., outcomes of two die rolls X and Y – interested in $E[X+Y]$
- Is it true that $E[X+Y] = E[X] + E[Y]$?

Sum of Expectations

- **Example.** I purchased two electronic components, each of which may be defective or acceptable
- $(d, d), (d, a), (a, d), (a, a)$ – probabilities 0.09, 0.2, 0.22 and 0.49 respectively
- $X = 1$ if first component is acceptable; 0 otherwise
- $Y = 1$ if second component is acceptable; 0 otherwise
- $X+Y$ = no. of acceptable components
- X and Y are *not* independent

$$X \times Y = \# \text{ acceptable components} = 0, 1 \text{ or } 2$$
$$x = \begin{cases} 1 & \text{if 1st component is acceptable;} \\ 0 & \text{otherwise} \end{cases}$$
$$f = \{ \dots - 2^k \dots \}$$

$$E[X + Y] = E[X] + E[Y] ?$$

$X + Y$ takes values 0, 1, 2

pmf $p(0) = 0.09$

$$p(1) = 0.2 + 0.22 = 0.42$$

$$p(2) = 0.49$$

$$\begin{aligned} E[X + Y] &= 0 \times 0.09 + 1 \times 0.42 + 2 \times 0.49 \\ &= 1.4 \end{aligned}$$

X takes values $0, 1$

$$\begin{aligned} \text{pmf: } p(0) &= P((d,d)) + P((d,a)) = 0.09 + 0.2 \\ &= 0.29 \\ p(1) &= 0.71 \end{aligned} \quad \begin{aligned} E(X) &= 0 \times 0.29 + 1 \times 0.71 \\ &= 0.71 \end{aligned}$$

Y takes values $0, 1$

$$\begin{aligned} \text{pmf: } p(0) &= 0.09 + 0.22 \\ &= 0.31 \\ p(1) &= 0.69 \end{aligned} \quad \begin{aligned} E(Y) &= 0.69 \\ \text{So } E(X) + E(Y) &= 1.4 \end{aligned}$$

Sum of Expectations

- **Example.** I purchased two electronic components, each of which may be defective or acceptable
- $(d, d), (d, a), (a, d), (a, a)$ – probabilities 0.09, 0.2, 0.22 and 0.49 respectively
- $X = 1$ if first component is acceptable; 0 otherwise
- $Y = 1$ if second component is acceptable; 0 otherwise
- $X+Y$ = no. of acceptable components
- X and Y are *not* independent
- $P(X=1) = 0.22 + 0.49 = 0.71$; $E[X] = 1 * P(X=1) = 0.71$
- $P(Y=1) = 0.2 + 0.49 = 0.69$; $E[Y] = 0.69$
- $E[X+Y] = 0 * P(X+Y=0) + 1 * P(X+Y=1) + 2 * P(X+Y=2) = 1 * (0.2 + 0.22) + 2 * 0.49 = 1.4$

Sum of Expectations

- $E[X+Y] = E[X] + E[Y]$ is true in general
- $Z = X + Y$

$$\mathbb{E}(X + Y) = \mathbb{E}(Z) = \sum_z z \mathbb{P}(Z = z) = \sum_z z \mathbb{P}(X + Y = z)$$

$$\mathbb{P}(X + Y = z) = \sum_{x,y:x+y=z} \mathbb{P}(X = x, Y = y)$$

Sum of Expectations

$$\begin{aligned}\mathbb{E}[Z] &= \sum_z z \sum_{x,y:x+y=z} \mathbb{P}(X = x, Y = y) \\&= \sum_z \sum_{x,y:x+y=z} z \mathbb{P}(X = x, Y = y) \\&= \sum_{x,y} (x + y) \mathbb{P}(X = x, Y = y) \\&= \sum_{x,y} x \mathbb{P}(X = x, Y = y) + \sum_{x,y} y \mathbb{P}(X = x, Y = y) \\&= \sum_x x \sum_y \mathbb{P}(X = x, Y = y) + \sum_y y \sum_x \mathbb{P}(X = x, Y = y) \\&= \sum_x x \mathbb{P}(X = x) + \sum_y y \mathbb{P}(Y = y) \\&= \mathbb{E}[X] + \mathbb{E}[Y]\end{aligned}$$

Expectation of Random Variables

- Can generalize the argument to show that for discrete random variables $X_1, X_2, \dots, X_n$, $E[X_1 + X_2 + \dots + X_n] = E[X_1] + \dots + E[X_n]$
- **Example.** A construction firm has recently sent in bids for 3 jobs worth (in profits) 10, 20, and 40 thousand dollars respectively. If the probabilities of winning the jobs are respectively 0.2, 0.8 and 0.3, what is the firm's expected total profit?

$X = \text{total profit}$

X takes values 0, 10, 20, 30, 40, 50, 60, 70,

pmf: $p(0) = (1 - 0.2)(1 - 0.8)(1 - 0.3)$

$$p(10) = 0.2 \times (1 - 0.8)(1 - 0.3) - \dots$$

X = total profit

$$= X_1 + X_2 + X_3$$

where X_1 = profit from job 1

X_2 = ... 2

X_3 = ... 3

$$E(X) = E(X_1) + E(X_2) + E(X_3)$$

$$X_1 = \begin{cases} 10 & \text{w.p. } 0.2 \\ 0 & \text{w.p. } 0.8 \end{cases}$$

$$X_2 = \begin{cases} 20 & \text{w.p. } 0.8 \\ 0 & \text{w.p. } 0.2 \end{cases}$$

$$X_3 = \begin{cases} 40 & \text{w.p. } 0.3 \\ 0 & \text{w.p. } 0.7 \end{cases}$$

$$E(X_1) = (0 \times 0.2 + 0.8 \times 0) = 2$$

$$E(X_2) = 20 \times 0.8 = 16, \quad E(X_3) = 40 \times 0.3 = 12$$

$$\begin{aligned} E(x) &= 2 + 16 + 12 \\ &= 30 \end{aligned}$$

Expectation of Random Variables

- Can generalize the argument to show that for discrete random variables $X_1, X_2, \dots, X_n$, $E[X_1 + X_2 + \dots + X_n] = E[X_1] + \dots + E[X_n]$
- **Example.** A construction firm has recently sent in bids for 3 jobs worth (in profits) 10, 20, and 40 thousand dollars respectively. If the probabilities of winning the jobs are respectively 0.2, 0.8 and 0.3, what is the firm's expected total profit?
- **Solution.** Let X_i , $i = 1, 2, 3$ denote the firm's profit from job i , then total profit $= X_1 + X_2 + X_3$. So $E[\text{total profit}] = E[X_1 + X_2 + X_3] = E[X_1] + E[X_2] + E[X_3]$.
- $E[X_1] = 10 \times 0.2 + 0 \times 0.8 = 2$; $E[X_2] = 16$ and $E[X_3] = 12$
- $E[X_1 + X_2 + X_3] = 2 + 16 + 12 = 30$

Expectation of Random Variables

- **Example.** What is the expected value of the number of times that in a thoroughly shuffled deck of 52 cards two adjacent cards are of the same rank (two aces, two kings, etc)?

$X =$ # same rank as previous card

$$X = X_1 + X_2 + \dots + X_{51}$$

where

$$X_i = \begin{cases} 1 & \text{if } i\text{-th and } (i+1)\text{-th} \\ & \text{cards have same rank} \\ 0 & \text{otherwise} \end{cases}$$

$$E[X_i] = 1 \times P(X_i = 1) + 0 \times P(X_i = 0) \\ = P(X_i = 1)$$

$$P(X_1 = 1) = P(\text{1}^{\text{st}} \text{ and } 2^{\text{nd}} \text{ cards have same rank})$$

$$= \frac{3}{51} \quad // \quad \frac{3}{51}$$

$$P(X_2 = 1) = P(\text{2}^{\text{nd}} \text{ and } 3^{\text{rd}} \text{ cards have same rank})$$

$$P(X_i = 1) = P(i\text{-th and } (i+1)\text{-th cards have same rank})$$

$$= \frac{3}{51}$$

$$E(X) = E(X_1) + \dots + E(X_{51})$$

$$= \frac{3}{51} + \dots + \frac{3}{51} = 3$$

Pick 1st and 2nd cards out

$P(1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ cards have same rank})$

Pick 10th and 11th cards out

$P(10^{\text{th}} \text{ and } 11^{\text{th}} \text{ cards have same rank})$

A random variable, e.g., X_i , that takes values 0 or 1 is called an indicator variable.

$$E[X_i] = P(X_i = 1)$$

Expectation of Random Variables

- **Example.** What is the expected value of the number of times that in a thoroughly shuffled deck of 52 cards two adjacent cards are of the same rank (two aces, two kings, etc)?
- **Solution.** Let X_i be 1 if the cards in position i and $i+1$ are of the same rank, and 0 otherwise. Then the quantity of interest is $E[X_1 + X_2 + \dots + X_{51}] = E[X_1] + \dots + E[X_{51}]$.
- $E[X_1] = 1 \times 3/51 + 0 \times 48/51 = 3/51$
- Similarly, $E[X_i] = 3/51$
- $E[X_1 + X_2 + \dots + X_{51}] = 3/51 \times 51 = 3$

Expectation of Random Variables

- **Example.** A secretary has typed N letters along with their respective envelopes. The envelopes get mixed up when they fall on the floor. If the letters are placed in the mixed-up envelopes in a completely random manner, what is the expected number of letters that are placed in the correct envelopes?

$X = \# \text{ letters placed in correct envelopes}$

$$X = X_1 + \dots + X_N$$

where

$$X_i = \begin{cases} 1 & \text{if letter } i \text{ is placed in correct envelope} \\ 0 & \text{otherwise} \end{cases}$$

Expectation of Random Variables

- **Example.** A secretary has typed N letters along with their respective envelopes. The envelopes get mixed up when they fall on the floor. If the letters are placed in the mixed-up envelopes in a completely random manner, what is the expected number of letters that are placed in the correct envelopes?
- **Solution.** Let $X_i = 1$ if the i^{th} letter is placed in the correct envelope and 0 otherwise, $i = 1, 2, \dots, N$. Let X = total number of correctly placed letters
- $X = X_1 + X_2 + \dots + X_N$
- $E[X_i] = 1P(X_i=1) + 0P(X_i=0) = P(X_i=1) = 1/N$
- $E[X] = 1$

$$E[X_i] = P(X_i = 1)$$

$$= P(\text{letter } i \text{ placed in correct envelope})$$

$$= \frac{1}{n}$$

$$E[X] = \frac{1}{n} + \dots + \frac{1}{n} = 1$$